

Question				Marking details		Marks available																																			
						AO1	AO2	AO3	Total	Maths	Prac																														
4.	(a)	(i)		- Woodland: brown and unbanded snails (1) - Meadow: random/more variation/slightly more yellow/ owtte (1) - Hedge: (are any colour and/but) mainly banded (1)				3	3		3																														
		(ii)		Due to {camouflage/ temperature/ { selective predation/ or description of};				1	1																																
	(b)			<table><tr><td>Habitat</td><td>O</td><td>E</td><td>O-E</td><td>(O-E)²</td><td>$\frac{(O-E)^2}{E}$</td></tr><tr><td>Wood</td><td>28</td><td>11</td><td>17</td><td>289</td><td>26.27</td></tr><tr><td>Hedge</td><td>2</td><td>11</td><td>-9</td><td>81</td><td>7.36</td></tr><tr><td>Meadow</td><td>3</td><td>11</td><td>-8</td><td>64</td><td>5.82</td></tr><tr><td></td><td>Total 33</td><td></td><td></td><td></td><td>Σ=39.45</td></tr></table> Σ=39.45 = 3 marks If incorrect award 2marks for 39.5 39.44 Correct (O-E)² column If incorrect award 1 marks for Correct Expected column		Habitat	O	E	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$	Wood	28	11	17	289	26.27	Hedge	2	11	-9	81	7.36	Meadow	3	11	-8	64	5.82		Total 33				Σ=39.45		3		3	3	3
Habitat	O	E	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$																																				
Wood	28	11	17	289	26.27																																				
Hedge	2	11	-9	81	7.36																																				
Meadow	3	11	-8	64	5.82																																				
	Total 33				Σ=39.45																																				
	(c)			Reject the null hypothesis (1) {39.45/ calculated value} is much greater than the critical value of 5.99 (1) Accept 5.99 circled on table if not mentioned ecf (b) This means that the difference between O and E is significant / therefore the results are not due to chance/ Some other factor is affecting their high numbers in the woodland (1)				3	3	3	3																														
				Question 4 total		0	3	7	10	5	9																														